

Eq. (5) gives the detectability duration in terms of the mean duty cycle and time in the observation window:

$$L_{\text{eff}} = D_c T_{\text{span}}. \quad (1)$$

Some technosignatures may have a lifetime exactly equal to the the civilization's lifetime, but others may persist for some time after collapse. For technosignature i , the detectability duration is given as

$$L_i = D_c T_{\text{span}} + \min[T_i, (1 - D_c) T_{\text{span}}] \delta_i, \quad (2)$$

where T_i is the lifetime of technosignature i and δ_i is a flag that is 1 if technosignature i is present in the scenario and 0 otherwise.

The probability that technosignature i could be observed (assuming an appropriate observatory) is

$$p_i = \frac{L_i}{T_{\text{span}}} = D_c + \min\left[\frac{T_i}{T_{\text{span}}}, 1 - D_c\right] \delta_i, \quad (3)$$

These probabilities are shown in the figure above for four example technosignatures, with a 1000-year time span. These probabilities should be interpreted as the probability of finding a given technosignature for remote observations of Earth in a given scenario, assuming the use of a capable observatory and sufficient observation time. Nitrogen dioxide (NO_2) has a short lifetime of hours to days, so the probability is about equal to the duty cycle for scenarios in which nitrogen dioxide is present. The lifetimes of CFC-11 is 55 yr, and the lifetime of CFC-12 is 140 yr, which could be observed in five of the scenarios. CFC-12 has higher observation probabilities due to its longer lifetime. Carbon tetrafluoride (CF_4) has a lifetime of 1000 yr or longer, observable in one of the scenarios.

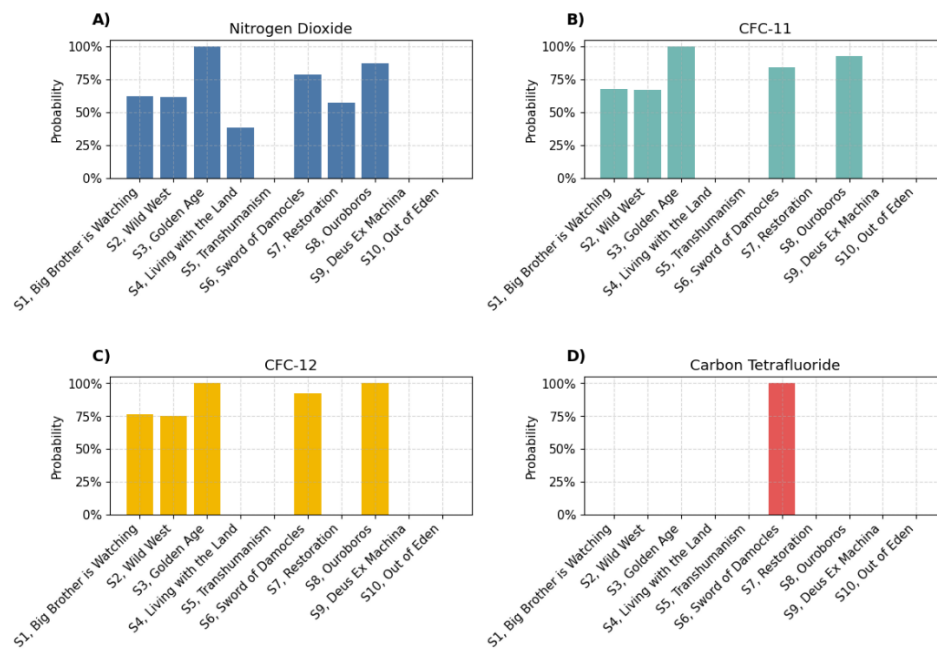


Figure 1: The probability that a technosignature could be observed, taking into account the duty cycle for each scenario and the lifetime of each technosignature.